

Healthcare Data Hub — data mart manifest

Entity relationships, primary keys, business keys and foreign keys for all 34 mart tables.
Generated 2026-09-15 from the shipped data.

Synthetic data. Every record described here is wholly fabricated. No row corresponds to any real person, provider, facility or payer. Because the data is fabricated rather than de-identified, HIPAA de-identification standards do not apply and are not claimed — this dataset is not “HIPAA compliant”, a concept that does not apply to invented records.

34 MART TABLES	9.6M ROWS	56 FOREIGN KEYS
30 of 30 DECLARED PRIMARY KEYS UNIQUE	0 FOREIGN KEYS WITH ORPHANS	0 BUSINESS KEY VIOLATIONS

4 of the 34 tables are bridges or evaluation tables with no surrogate key by design; they are keyed on their business key, which is stated and verified for each. Every one of the 30 declared primary keys is unique and non-null.

How to read this

Three kinds of key appear against every table, and the distinction between the first two is the single most load-bearing idea in the document.

KEY	WHAT IT IS	WHAT IT IS FOR
PK	The surrogate primary key. One column, an integer, meaningless outside this mart.	Joining. It is unique <i>by construction</i> , which is exactly why testing it proves nothing about data quality.
BK	The business key: the natural grain of the table, stated in the columns a source system would recognise.	De-duplication and grain assertions. This is the key a uniqueness test must run on. A re-driven extract duplicated 2,840 claim lines in this dataset and every one carried a distinct surrogate key, so a primary-key test passed while the grain was broken.
FK	Foreign key, with its parent table and column.	Joining, and knowing what a join would drop. Null rates are stated because several are null <i>by design</i> .

Every figure was measured by `Build/verify_keys.py` against the files in `Mart/`. Nothing here is transcribed.

Layers

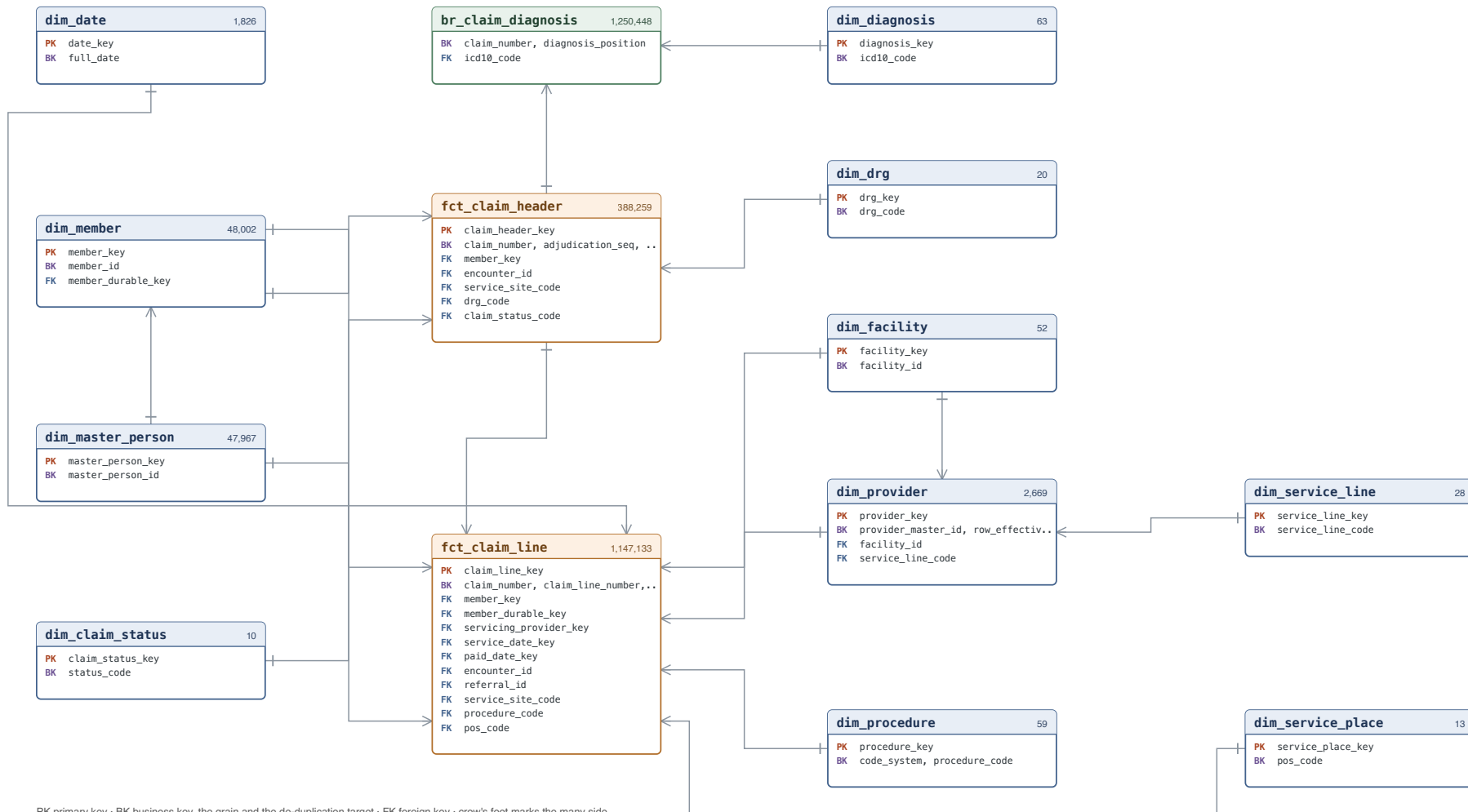
Five source systems land as gzipped CSV, a conformed and dimensional layer is built from them, and one wide view sits on top for self-service. The mart tables in this document are the middle and top of that stack.

LAYER	LOCATION	CONTENTS
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Source	Source Data/	Five systems as landed: string-typed, defects intact. The shippable artifact.
Mart	Mart/	Conformed dimensions, facts, bridges and crosswalks. Documented here.
Serving	Mart/vw_claim_line_enriched.csv.gz	One row per claim line, no bridges, safe to sum. Start here.

Claims star — the money, and everything it hangs off

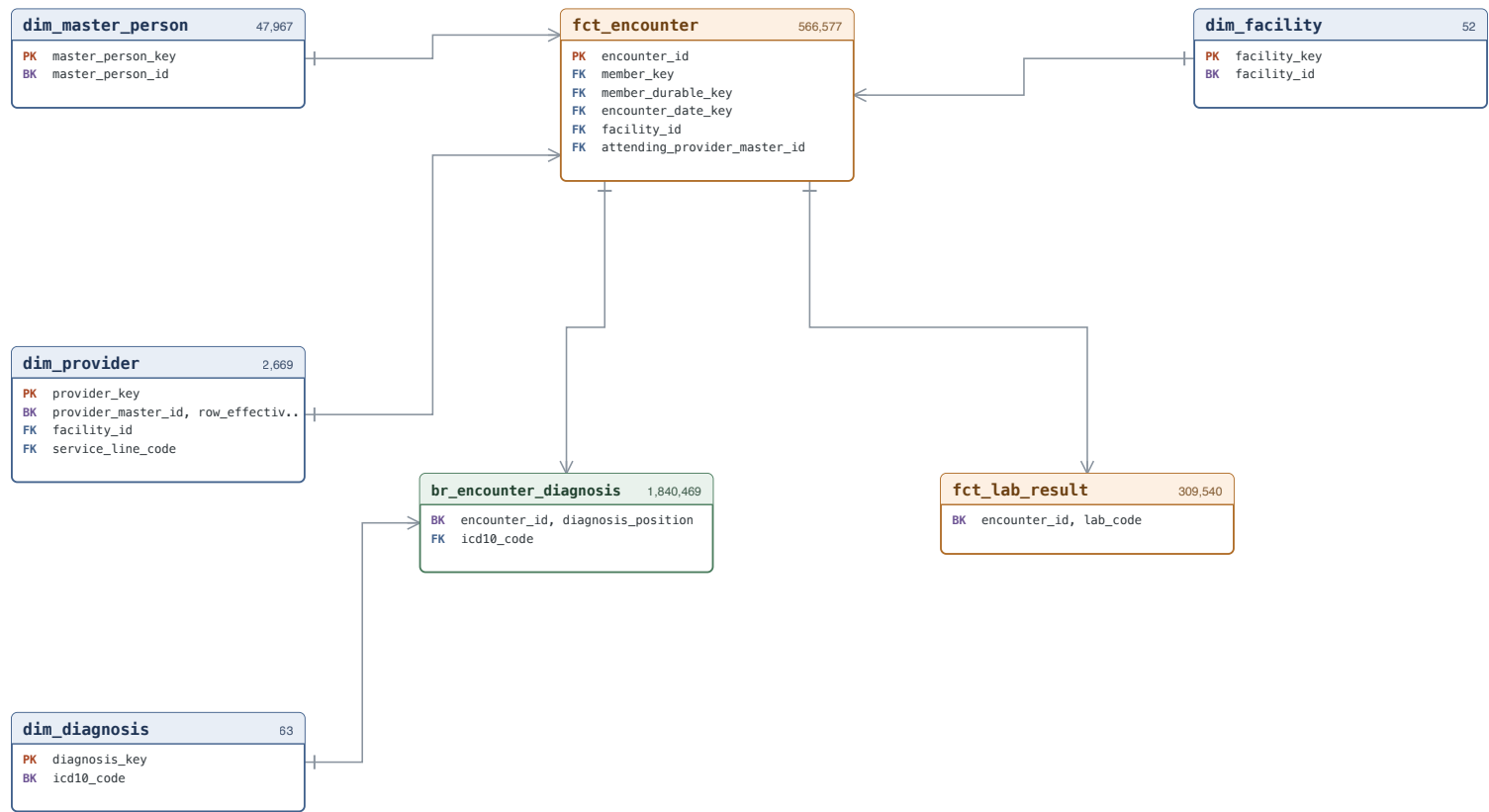
Claim lines de-duplicate on the BUSINESS key, never on claim_line_key: a re-driven extract duplicated 2,840 lines and every one carried a distinct surrogate. DRG, length of stay and the claim total live only on the header, so a DRG-by-procedure question is forced through that join, and br_claim_diagnosis is at HEADER grain so joining a line to it fans out.



PK primary key · BK business key, the grain and the de-duplication target · FK foreign key · crow's foot marks the many side

Clinical — the EHR side, keyed by MRN

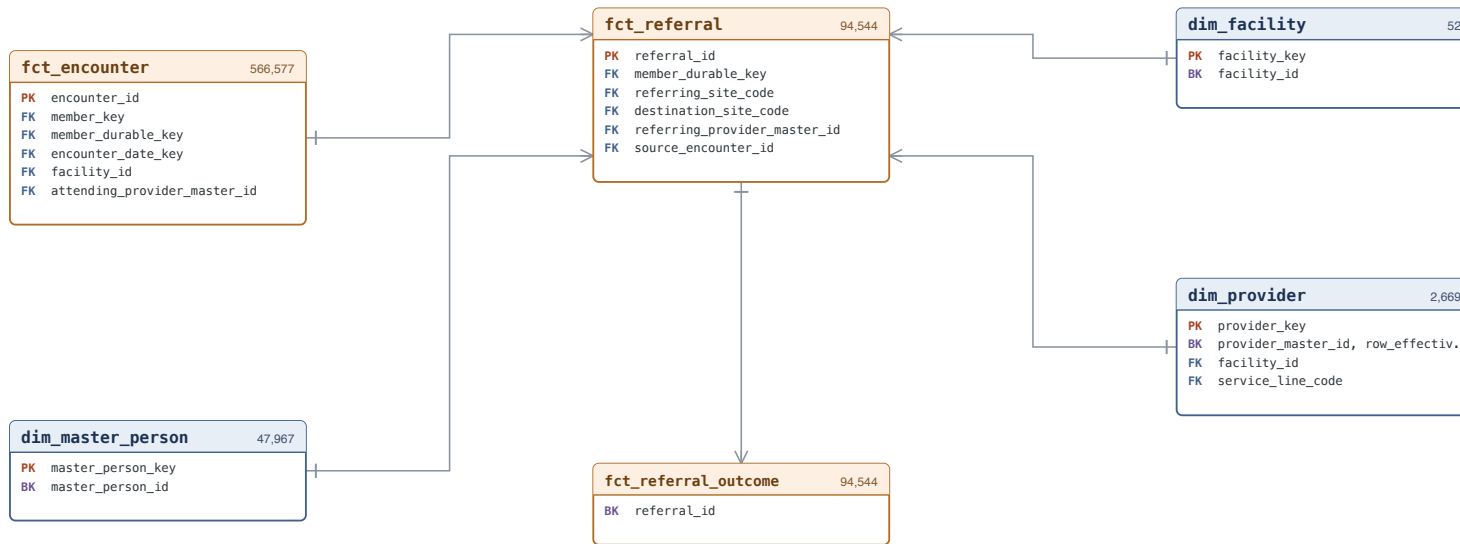
The EHR keys patients by MRN, so nothing here joins to claims directly — it has to go through xwalk_patient (next diagram), and that hop loses roughly 3% of records non-randomly. br_encounter_diagnosis averages about 3.25 rows per encounter and will multiply any measure summed after it.



PK primary key · BK business key, the grain and the de-duplication target · FK foreign key · crow's foot marks the many side

Referral chain — intent, and whether care actually happened

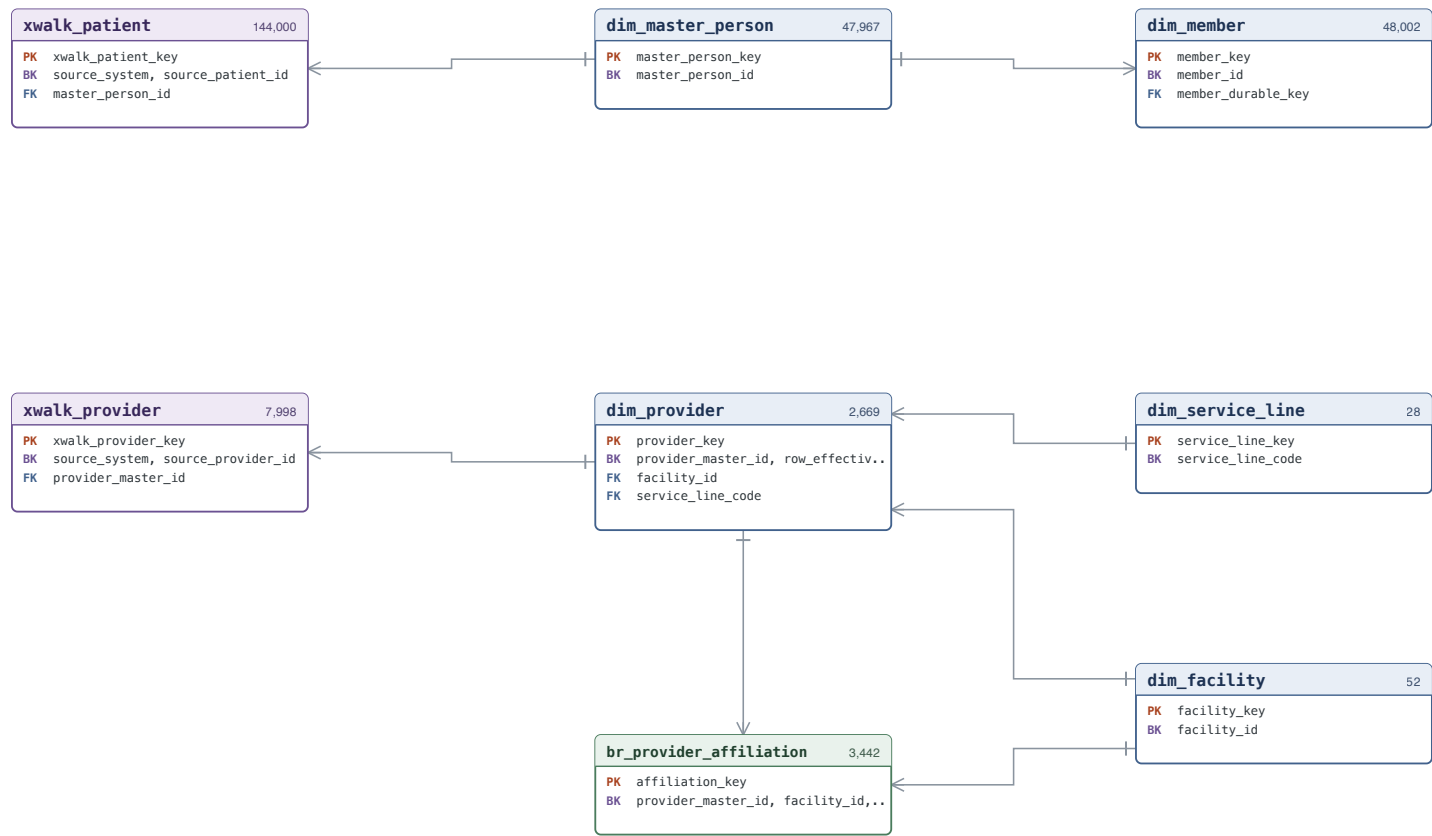
fct_referral records intent and what the EHR directory BELIEVED about the destination at order time — never the fact. fct_referral_outcome is where a referral is tested against a real confirming event, a claim or a completed appointment, rather than against its own status field.



PK primary key · BK business key, the grain and the de-duplication target · FK foreign key · crow's foot marks the many side

Identity resolution and provider structure

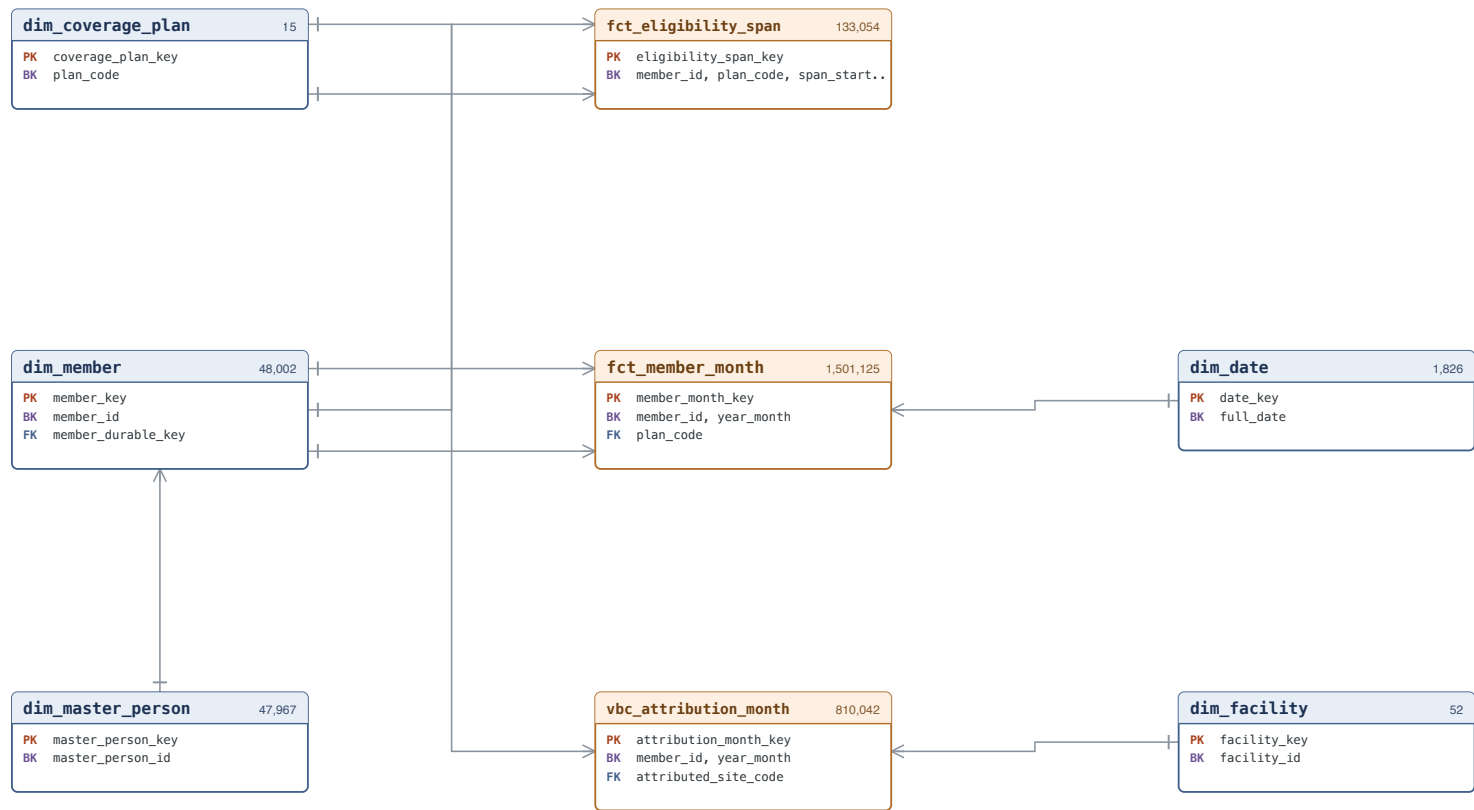
Three patient key spaces and three provider key spaces resolve here. The crosswalk is imperfect in both directions on purpose.
br_provider_affiliation is the fan-out hazard: 22% of providers hold two or more concurrent affiliations, and allocation_pct sums to exactly 1.0 per provider-span for anyone who wants a weighted answer.



PK primary key · BK business key, the grain and the de-duplication target · FK foreign key · crow's foot marks the many side

Membership, coverage and attribution — where the denominator comes from

fct_member_month is the only sanctioned PMPM denominator, built as the gaps-and-islands union of coverage. fct_eligibility_span contains deliberate overlaps, so summing span lengths double counts. The attribution roster is restated retroactively, so compare versions rather than trusting the current roster.



PK primary key · BK business key, the grain and the de-duplication target · FK foreign key · crow's foot marks the many side

Table reference

Grain, keys and relationships for every table, grouped as the diagrams group them. nullable by design is called out wherever a null foreign key is expected rather than a defect.

CONFORMED DIMENSIONS

dim_date

One row = one calendar day · 1,826 rows · 23 columns

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	date_key	—	unique, no nulls
BK	full_date	—	unique

Role-played as service, incurred and paid date. Carries claims_runout_complete_flag, the guardrail for any service-date trend.

dim_member

One row = one member · 48,002 rows · 17 columns · includes 2 sentinel row(s)

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	member_key	—	unique, no nulls
BK	member_id	—	unique
FK	member_durable_key	dim_master_person.master_person_id	no orphans · the resolved person behind this member

Carries the -1 Unknown and -2 Not Applicable members. Type 1 on demographics.

dim_master_person

One row = one resolved person · 47,967 rows · 12 columns

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	master_person_key	—	unique, no nulls
BK	master_person_id	—	unique

Resolution quality on every row. ehr_records_resolved > 1 means an over-match: two real people on one identity.

dim_provider

One row = one provider per tracked-attribute change (type 2) · 2,669 rows · 20 columns · includes 2 sentinel row(s)

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	provider_key	—	unique, no nulls
BK	provider_master_id, row_effective_date	—	unique
FK	facility_id	dim_facility.facility_id	no orphans · 0.07% null
FK	service_line_code	dim_service_line.service_line_code	no orphans · 0.07% null

Type 2. provider_master_id is the durable key; provider_key is point-in-time correct. Role-played four ways: referring, performing, billing, attributed PCP.

dim_provider_current

One row = one provider (current row only) · 2,666 rows · 20 columns

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	provider_key	—	unique, no nulls
BK	provider_master_id	—	unique
FK	facility_id	dim_facility.facility_id	no orphans

The convenience view. Using it reassigns a provider's whole history to their current facility - which is the point of the as-of-join lesson.

dim_facility

One row = one site of care · 52 rows · 17 columns

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	facility_key	—	unique, no nulls
BK	facility_id	—	unique

site_code is the join key used throughout the facts. Carries performs_msk_surgery and annual_msk_case_capacity.

dim_coverage_plan

One row = one payer x product x plan x benefit year · 15 rows · 14 columns

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	coverage_plan_key	—	unique, no nulls
BK	plan_code	—	unique

Payer is an ATTRIBUTE here, not its own dimension: every attribute analysts slice by lives at plan grain.

dim_diagnosis

One row = one ICD-10-CM code · 63 rows · 12 columns

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	diagnosis_key	—	unique, no nulls
BK	icd10_code	—	unique

sex_restriction and max_age drive the clinical plausibility assertions.

dim_procedure

One row = one procedure code, all code systems · 59 rows · 11 columns

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	procedure_key	—	unique, no nulls
BK	code_system, procedure_code	—	unique

One dimension with code_system as discriminator. Splitting CPT from ICD-10-PCS would force a UNION on every procedure query.

dim_service_place

One row = one place-of-service code · 13 rows · 5 columns

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	service_place_key	—	unique, no nulls
BK	pos_code	—	unique

dim_drg

One row = one MS-DRG · 20 rows · 7 columns

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	drg_key	—	unique, no nulls
BK	drg_code	—	unique

Joins to the claim HEADER only. DRG is a header attribute.

dim_claim_status

One row = one adjudication status · 10 rows · 8 columns

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	claim_status_key	—	unique, no nulls
BK	status_code	—	unique

CO-45 is deliberately absent: the contractual write-off is an AMOUNT on the line, not a denial status.

dim_service_line

One row = one service line · 28 rows · 5 columns

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	service_line_key	—	unique, no nulls
BK	service_line_code	—	unique

CLAIMS FACTS

fct_claim_header

One row = one claim x one adjudication version · 388,259 rows · 39 columns

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	claim_header_key	—	unique, no nulls
BK	claim_number, adjudication_seq, net_sign	—	unique
FK	member_key	dim_member.member_key	no orphans
FK	encounter_id	fct_encounter.encounter_id	no orphans · 30.03% null · NULLABLE by design: ~24% of claims have no Northlake encounter
FK	service_site_code	dim_facility.site_code	no orphans
FK	drg_code	dim_drg.drg_code	no orphans · 97.67% null · inpatient facility claims only
FK	claim_status_code	dim_claim_status.status_code	no orphans

DRG, admit/discharge, length of stay, disposition and the claim total live ONLY here, so a DRG-by-procedure question is forced through the header-to-line join.

fct_claim_line

One row = one claim x one line x one adjudication version · 1,147,133 rows · 55 columns

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	claim_line_key	—	unique, no nulls
BK	claim_number, claim_line_number, adjudication_seq, net_sign	—	unique
FK	member_key	dim_member.member_key	no orphans
FK	member_durable_key	dim_master_person.master_person_id	no orphans
FK	servicing_provider_key	dim_provider.provider_key	no orphans · resolves to -1 where credentialing has no record
FK	service_date_key	dim_date.date_key	no orphans
FK	paid_date_key	dim_date.date_key	no orphans
FK	claim_number	fct_claim_header.claim_number	no orphans
FK	encounter_id	fct_encounter.encounter_id	no orphans · 27.05% null · nullable
FK	referral_id	fct_referral.referral_id	no orphans · 72.95% null · populated on referral episodes and consults only
FK	service_site_code	dim_facility.site_code	no orphans
FK	procedure_code	dim_procedure.procedure_code	no orphans
FK	pos_code	dim_service_place.pos_code	no orphans

The de-duplication target. The SURROGATE claim_line_key is unique even on the raw file, so a primary-key uniqueness test passes while the business key is violated.

br_claim_diagnosis

One row = one claim header x one diagnosis position (1-12) · 1,250,448 rows · 5 columns

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	none — bridge, keyed on its business key		—
BK	claim_number, diagnosis_position	—	unique
FK	claim_number	fct_claim_header.claim_number	no orphans
FK	icd10_code	dim_diagnosis.icd10_code	no orphans

HEADER grain, exactly like an 837. Claim LINES carry pointers into this list rather than their own codes. Joining a line to this bridge FANS OUT - that is correct behavior, not a bug.

CLINICAL FACTS

fct_encounter

One row = one EHR encounter · 566,577 rows · 24 columns

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	encounter_id	—	unique, no nulls
BK	encounter_id	—	unique
FK	member_key	dim_member.member_key	no orphans · resolves to -1 where the crosswalk failed
FK	member_durable_key	dim_master_person.master_person_id	no orphans · 3.07% null
FK	encounter_date_key	dim_date.date_key	no orphans
FK	facility_id	dim_facility.facility_id	no orphans · resolved through the facility crosswalk
FK	attending_provider_master_id	dim_provider.provider_master_id	no orphans

The three-hop join to claims lives here, and it loses the unmatched share NON-randomly.

br_encounter_diagnosis

One row = one encounter x one diagnosis x position · 1,840,469 rows · 5 columns

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	none — bridge, keyed on its business key		—
BK	encounter_id, diagnosis_position	—	unique
FK	encounter_id	fct_encounter.encounter_id	no orphans
FK	icd10_code	dim_diagnosis.icd10_code	no orphans

Mean ~3.25 diagnoses per encounter. No pre-built cost allocation ships: deciding how to allocate is the lesson.

fct_lab_result

One row = one resulted lab component per encounter · 309,540 rows · 12 columns

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	none — bridge, keyed on its business key		—
BK	encounter_id, lab_code	—	unique
FK	encounter_id	fct_encounter.encounter_id	no orphans

result_status = NO_STRUCTURED_RESULT where an external lab returned nothing parseable - one half of the HbA1c gap.

REFERRAL FACTS

fct_referral

One row = one referral order · 94,544 rows · 17 columns

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	referral_id	—	unique, no nulls
BK	referral_id	—	unique
FK	member_durable_key	dim_master_person.master_person_id	no orphans · 3.2% null
FK	referring_site_code	dim_facility.site_code	no orphans
FK	destination_site_code	dim_facility.site_code	no orphans
FK	referring_provider_master_id	dim_provider.provider_master_id	no orphans · 4.0% null · nullable: ~4% carry no referring provider
FK	source_encounter_id	fct_encounter.encounter_id	no orphans

days_to_closure and closure_actor_type are the fingerprints of the auto-close rule. destination_status_per_ehr_directory is the STALE belief, never the fact.

fct_referral_outcome

One row = one referral, evaluated against confirming events · 94,544 rows · 13 columns

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	none — bridge, keyed on its business key		—
BK	referral_id	—	unique
FK	referral_id	fct_referral.referral_id	no orphans

An auditable bridge with match_rule_version on every row. confirm_source records whether a CLAIM, an APPOINTMENT, BOTH or NONE confirmed the referral - two independent paths, so the finding survives a challenge to either join.

MEMBERSHIP AND CONTRACT

fct_eligibility_span

One row = one member x plan x contiguous coverage span · 133,054 rows · 14 columns

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	eligibility_span_key	—	unique, no nulls
BK	member_id, plan_code, span_start_date, coverage_type	—	unique
FK	member_id	dim_member.member_id	no orphans
FK	plan_code	dim_coverage_plan.plan_code	no orphans

NOT a denominator. Contains deliberate overlaps; summing span lengths double counts every one of them. coverage_type is part of the grain: a COBRA span legitimately starts on the same day as the active coverage it runs alongside.

fct_member_month

One row = one member x one year-month with any coverage · 1,501,125 rows · 16 columns

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	member_month_key	—	unique, no nulls
BK	member_id, year_month	—	unique
FK	member_id	dim_member.member_id	no orphans
FK	year_month	dim_date.year_month	no orphans
FK	plan_code	dim_coverage_plan.plan_code	no orphans

The ONLY sanctioned PMPM denominator. Built as the gaps-and-islands UNION of coverage, not the sum of spans.

vbc_attribution_month

One row = one member x one month, current roster version · 810,042 rows · 12 columns

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	attribution_month_key	—	unique, no nulls
BK	member_id, year_month	—	unique
FK	member_id	dim_member.member_id	no orphans
FK	attributed_site_code	dim_facility.site_code	no orphans

Restated retroactively. Compare roster VERSIONS, not just the current roster, or a denominator change reads as performance.

vbc_attribution_restatement

One row = one member x month whose attribution status changed, x the roster version that changed it · 784 rows · 14 columns

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	restatement_key	—	unique, no nulls
BK	member_id, year_month, as_of_version	—	unique
FK	member_id	dim_member.member_id	no orphans
FK	as_of_version	vbc_roster_version.as_of_version	no orphans
FK	attributed_site_code	dim_facility.site_code	no orphans

The delta, not a full versioned history. Roster at version V = current roster, minus additions applied after V, plus terminations applied after V. Carries member_months and risk_score so that reconstruction needs no other table. Changes run in BOTH directions - new_status is RETRO_TERMINATED or ATTRIBUTED.

vbc_roster_version

One row = one monthly roster production run · 6 rows · 11 columns

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	roster_version_key	—	unique, no nulls
BK	as_of_version	—	unique

Six rows. Exists so an as-of control has something to bind to and so 'which version is current' is stated in the data rather than left to a MAX().

vbc_benchmark

One row = one month x line of business · 72 rows · 9 columns

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	benchmark_key	—	unique, no nulls
BK	year_month, line_of_business	—	unique
FK	year_month	dim_date.year_month	no orphans

Calibrated against the ATTRIBUTED cohort, not the whole book, and risk-adjusted through a score normalized to a book mean of 1.0 within line of business.

vbc_contract_terms

One row = one contract x performance year · 6 rows · 15 columns

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	contract_term_key	—	unique, no nulls
BK	contract_id, performance_year	—	unique

Minimum savings rate, shared savings and loss rates, quality gate, and the high-cost truncation threshold. Without these a workbook can show PMPM moved and cannot show the cheque moved.

SETTLEMENT AND COMPLETENESS

fct_member_year_cost

One row = one member x performance year · 53,501 rows · 13 columns

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	member_year_cost_key	—	unique, no nulls
BK	member_id, performance_year	—	unique
FK	member_id	dim_member.member_id	no orphans
FK	member_durable_key	dim_master_person.master_person_id	no orphans

High-cost truncation is an ANNUAL, MEMBER-LEVEL cap. Applying it per claim line caps nothing, because no single line reaches the threshold. fct_claim_line.allowed_amount_truncated is this cap allocated back down pro rata and sums to it exactly.

fct_claims_lag_triangle

One row = one service month x lag month (0-12, tail lumped at 12) · 468 rows · 12 columns

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	lag_triangle_key	—	unique, no nulls
BK	service_year_month, lag_months	—	unique
FK	service_year_month	dim_date.year_month	no orphans

Dense: a lag with no claims is a zero row, not a missing one. is_observable is true only where the WHOLE development period had elapsed by the paid-through date - the end of the lag month, not its start. completion_factor_derived is chain ladder over observable cells and reproduces dim_date.claims_completeness_factor from the data alone.

BRIDGES AND CROSSWALKS

br_provider_affiliation

One row = one provider x facility x effective span · 3,442 rows · 11 columns

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	affiliation_key	—	unique, no nulls
BK	provider_master_id, facility_id, effective_date	—	unique
FK	provider_master_id	dim_provider.provider_master_id	no orphans
FK	facility_id	dim_facility.facility_id	no orphans

22% of providers hold two or more concurrent affiliations, so an unweighted join FANS OUT and overstates facility totals. allocation_pct sums to exactly 1.0 per provider-span.

xwalk_patient

One row = one source system x one source patient identifier · 144,000 rows · 9 columns

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	xwalk_patient_key	—	unique, no nulls
BK	source_system, source_patient_id	—	unique
FK	master_person_id	dim_master_person.master_person_id	no orphans · 2.08% null · NULL where the record never matched

Three patient key spaces. Imperfect in BOTH directions: under-match loses records non-randomly, over-match collapses two real people onto one identity.

xwalk_provider

One row = one source system x one source provider identifier · 7,998 rows · 10 columns

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	xwalk_provider_key	—	unique, no nulls
BK	source_system, source_provider_id	—	unique
FK	provider_master_id	dim_provider.provider_master_id	no orphans

Three provider key spaces: credentialed NPI, EHR internal id, and claim-side NPI.

SERVING LAYER

vw_claim_line_enriched

One row = one claim line (denormalized) · 1,147,133 rows · 48 columns

KEY	COLUMN(S)	REFERENCES	VERIFIED
PK	claim_line_key	—	unique, no nulls
BK	claim_number, claim_line_number, adjudication_seq, net_sign	—	unique
FK	member_key	dim_member.member_key	no orphans
FK	servicing_provider_key	dim_provider.provider_key	no orphans

Claim line joined to every type-1 dimension, NO bridges, one row per line, safe to sum. It is only safe BECAUSE the diagnosis bridges were kept out of it.

Join hazards

Six things in this mart will produce a confidently wrong number if joined carelessly. All six are deliberate.

HAZARD	WHAT GOES WRONG, AND THE CONTROL
Surrogate vs business key	A uniqueness test on <code>claim_line_key</code> passes on the raw file while 2,840 duplicate lines sit in it. Test the business key — <code>claim_number + claim_line_number + adjudication_seq + net_sign</code> — and de-duplicate on that.
Diagnosis bridges fan out	<code>br_claim_diagnosis</code> is at HEADER grain and <code>br_encounter_diagnosis</code> averages ~3.25 rows per encounter. Joining a fact to either multiplies rows, so any measure summed afterwards is inflated. Filter on <code>is_primary</code> , or aggregate the bridge before joining.
Provider affiliation fans out	22% of providers hold two or more concurrent facility affiliations, so an unweighted join to <code>br_provider_affiliation</code> overstates facility totals by roughly 29%. Use the <code>service_site_code</code> already on the fact, or weight by <code>allocation_pct</code> , which sums to exactly 1.0 per provider-span.
Spans are not a denominator	<code>fct_eligibility_span</code> contains deliberate overlaps. Summing span lengths double counts every one. Use <code>fct_member_month</code> , which is the gaps-and-islands union of coverage and the only sanctioned PMPM denominator.
Reversals and versions	Two nettings tie exactly: <code>SUM(paid_amount)</code> over every row, and the same sum WHERE <code>is_current_version</code> . Note there is no reversal exclusion in the second — orphan reversals are current and negative on purpose, so filtering them breaks the tie. Filtering <code>adjudication_seq = 1</code> or <code>paid_amount > 0</code> both overstate paid.
Claims are not admissions	One inpatient stay emits one institutional claim plus several professional ones, all sharing an <code>encounter_id</code> . Counting claims as admissions inflates volume several fold. Use <code>COUNT(DISTINCT encounter_id)</code> .

Reading the flat files

One practical trap, worth stating because it bit this document's own verification script before it was fixed. A key column containing nulls is inferred as a float when read back, so 600140 becomes 600140.0 and a string comparison against an integer parent key reports **100% orphans** on a perfectly sound relationship.

Affected key columns in this mart:

TABLE	COLUMN	FIX
<code>dim_provider</code>	<code>facility_id</code>	Read as a nullable integer, or cast both sides before joining.
<code>fct_claim_header</code>	<code>drg_code</code>	Read as a nullable integer, or cast both sides before joining.

In pandas: `dtype={'facility_id': 'Int64'}`. In Snowflake the DDL in SQL/ types these explicitly, so the problem does not arise once loaded.